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Article in International Journal of Work Organisation and Emotion · May 2019

DOI: 10.1504/IJWOE.2019.10023361

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Human emotion detection based on questionnaire and text analysis

Rajib Ghosh* and Ditipriya Sinha

Department of Computer Science and Engineering,
NIT Patna,
Patna, Bihar, India
Email: rajib.ghosh@nitp.ac.in
Email: ditipriya.cse@nitp.ac.in
*Corresponding author

Abstract: Human emotions have been described by some theorists as discrete and consistent responses to internal or external events which have a particular significance for the organism. Emotions are the topic of extensive research in the recent times. However, state of the art describes that most of the approaches on emotion detection have been designed on the basis of complex and costly approaches like physiological features, brain signals, etc. The present article presents a simple and cost-effective emotion detection model by combining questionnaire and text analysis-based approaches and then combining the probability scores of two different classifiers (support vector machine and artificial neural network) using Dempster-Shafer theory (DST). In the present work, DST has been used effectively in combining multiple information sources which provides incomplete, imprecise, and biased knowledge. Experimental results show that the proposed system outperforms all existing emotion detection systems available in the literature.

Keywords: human emotion detection; questionnaire; text analysis; support vector machine; SVM; artificial neural network; ANN; Dempster-Shafer theory; DST.

Reference to this paper should be made as follows: Ghosh, R. and Sinha, D. (2019) 'Human emotion detection based on questionnaire and text analysis', *Int. J. Work Organisation and Emotion*, Vol. 10, No. 1, pp.66–89.

Biographical notes: Rajib Ghosh is working as an Assistant Professor in Computer Science and Engineering Department at National Institute of Technology (NIT), Patna, India. He has more than 17 years of experiences of teaching in different engineering colleges. He completed his PhD (Computer Science and Engineering) from National Institute of Technology (NIT), Patna, India. He also holds MTech degree in Information Technology and BE degree in Computer Science and Engineering. His broader research domains are computer vision, pattern recognition and machine learning. His research areas of interest are document analysis and recognition, object detection, object tracking, human movement tracking, video surveillance, etc. His PhD work explored pattern recognition techniques for recognising online handwritten text of different Indic scripts such as Bengali, Devanagari, Telugu, Tamil, etc. He has published various research articles in different international journals and conferences of repute. He has also provided professional services as reviewers to many international conferences and SCI-indexed journals.

Ditipriya Sinha has received her PhD from the Department of Computer Science and Technology, Indian Institute of Engineering Science and Technology (IIST), Shibpur, India and Master of Technology from West Bengal University of Technology in the Department of Software Engineering. She was the silver medallist in her MTech degree. Currently, she is serving as an Assistant Professor at the Department of Computer Science and Engineering, National Institute of Technology, Patna. Earlier, she worked as an Assistant Professor at the Department of Computer Science and Engineering, Birla Institute of Technology, Mesra, India. Her areas of researches are mobile ad hoc network, wireless sensor network, scheduling algorithms and human emotion recognition.

1 Introduction

Human emotions play a vital role in our lives. Emotion can be broadly defined as “instinctive or intuitive feeling as distinguished from reasoning or knowledge”. They affect the ability of an individual to reason various situations and also govern their reaction to stimuli. Research explorations on emotion recognition have gained attraction in the recent past due to its various societal benefits. Emotion recognition finds its application in many areas such as medicine, law, marketing, e-learning, etc. Emotion identification is also considered as a key element for advanced human-computer interaction. Apart from human-computer interfaces, emotion recognition systems have applications in psychological counselling and in detecting criminal motives.

With the extensive researches in the fields such as Artificial Intelligence and Machine Learning, many works are being proposed to detect human emotion. Various approaches have already been proposed for emotion recognition. These include the use of various physiological features such as analysis of brain signals (Soleymani et al., 2016), heart rate (Nardelli et al., 2015; Valenza et al., 2012, 2014), pupil dilation (Partala and Surakka, 2003; Aracena et al., 2015; Lanata et al., 2011), skin conductance (Frantzidis et al., 2008) and facial expression recognitions (Soleymani et al., 2016; Happy and Routray, 2015; Chakraborty et al., 2009; Zhang and Ji, 2005). The approaches using physiological features require the assistance of expensive equipments to capture various physiological signals as well as facial expressions. So, these proposals are not cost effective. Apart from equipment-intensiveness, identification of human emotions from facial expressions is a challenging task for the machine due to following reasons: first, identification of human emotion from a blurred facial image is not a easy task and second, segmentation of a facial image into various regions is difficult if significant differences do not exist among different regions of the image. Apart from learning through psychological signals, some researchers proposed text-based model (Chaumartin, 2007; Tan and Zhang, 2008; Tokuhisa et al., 2008; Li et al., 2016) to detect emotions from subjective information based on blogs and other online social media such as Twitter (Li and Xu, 2014) by applying single classifier. But, to the best of our knowledge, no significant research works are available on emotion detection by combining multiple classifiers. Apart from text-based models, another kind of approach has been proposed through interpretation of situations/events (Binali et al., 2010) where the subject experiences that emotion. In this approach, the subjects are

asked to describe different events that make them experiencing different emotions, without necessarily mentioning the emotion itself.

The motivations of developing the present system are discussed below.

- 1 *Proper distribution of workplace loads:* a high 46% of the workforce in organisations in India suffer from some or the other form of stress, according to the latest data from Optum, a top provider of employee assistance programs to corporate (<https://economictimes.indiatimes.com/jobs/46-of-workforce-in-firms-in-india-suffer-from-some-or-theother-form-of-stress-data/articleshow/52696795.cms>). The proposed emotion detection system will help various organisations to analyse the stress level of its employee by detecting their emotional status and distributing the official workloads accordingly.
- 2 *Prevention of suicidal activity in educational institutions:* every hour, one student commits suicide in India, according to 2015 data (the latest available) from the National Crime Records Bureau (NCRB) (<https://www.hindustantimes.com/health-and-fitness/every-hour-one-student-commits-suicide-in-india/story-7UFFhSs6h1HNgrNO60FZ2O.html>). The proposed emotion detection system will help various educational institutions to assess the suicidal motive of its students by detecting their emotional status.
- 3 *Psychiatric counselling:* to assist psychiatrists in providing guidelines for psychiatric counselling to employees, students and other common people in general, to improve the overall public health service.

The research questions central to the present article are as follows:

- 1 How to achieve seamless integration of questionnaire and text analysis (blog-based) based features to produce flawless product on emotion detection?
- 2 How to automate decision in real time about the stress level of employees through detecting their emotional status using the combined features of questionnaire and blog data provided by them?
- 3 How to manage a huge volume of data generated from questionnaire-based approach, which are provided by the employees in workplace in response to various questions?
- 4 How to provide guidelines to psychiatrists of any organisation for psychiatric counselling of employees?
- 5 How to assist the head of any organisation to distribute workload among the employees analysing their emotional status?

In the proposed approach, behaviour of human beings have been analysed while experiencing a certain kind of emotion and based on that a questionnaire has been prepared. In this approach, the subjects have been asked some questions like ‘Do you wish to be alone?’ and similar physiological questions. So, rather than asking them about the emotion, we can detect it based on the answer of the subject to these questions. All the existing studies on text-based approach are based on interpretation of human events in which humans gain experience about a particular emotion followed by use of a single

classifier. We have proposed here the combination of questionnaire and text-analysis (blog-based) based approaches to generate the features and studied each feature vector using two different classifiers – support vector machine (SVM) and artificial neural network (ANN). Finally, the probability scores of SVM and ANN have been combined using Dempster-Shafer theory (DST) for enhancing the performance of the system. In the proposed system, DST has been developed effectively in combining multiple information sources which provides incomplete, imprecise and biased knowledge. The proposed approach is cost-effective and novel due to introduction of questionnaire-based approach along with the text analysis and combining the probability scores of two different classifiers – SVM and ANN applying DST. It is to be noted that, the proposed system outperforms all the existing emotion detection systems available in the literature.

The beneficiaries of the proposed system are employees of industries, young students and psychiatrists. Productivity of employees in various industries can be enhanced by overcoming depression due to work stress after detecting emotions using the proposed system. It will also enhance the economical growth of the industry and country. On the other hand, the proposed system will prevent students from taking suicidal attempts due to depression and encourage them to work towards the development of the country. Thus, educational institutions will be benefited from the proposed system. Similarly, the proposed system will assist psychiatrists to provide guidelines for psychiatric counselling in order to improve the public health service.

The rest of the paper is organised as follows. Section 2 describes the relevant and contextual works. In Section 3, the background of some theoretical concepts used is presented. Section 4 deals with the process of development of the datasets used in this work. The proposed approach of human emotion detection has been discussed in Section 5. The performance analysis of the proposed system is discussed in Section 6. Finally, we conclude with future possibilities of this work in Section 7.

2 Literature survey

2.1 Continuous emotion recognition

Wollmer et al. (2008) suggested throwing over the emotional classes in favour of dimensions and applied it on emotion recognition from speech. Nicolaou et al. (2011) used audiovisual modalities to detect valence and arousal on SEMAINE database (McKeown et al., 2012). In this work, support vector regression (SVR) and bidirectional long short-term memory recurrent neural networks (BLSTM-RNN) have been used to detect emotion continuously in time and dimensions. Nicolaou et al. (2012) also proposed a model for continuous emotion detection using an output-associative relevance vector machines (RVM) which smooth the RVM output. Although, in this work, the authors showed how it improved the performance of RVM for continuous emotion detection they did not compare its performance directly to the BLSTM recurrent neural network.

One of the major attempts in upgrading the state of the art in continuous emotion detection was the audio/visual emotion challenge (AVEC) 2012 (Schuller et al., 2012) which was proposed using SEMAINE database. SEMAINE database includes the audio-visual repercussion of participants recorded while interacting with the sensitive affective listeners (SAL) agents. The repercussion was continuously annotated on four

dimensions of valence, activation, power and expectation. The goal of the AVEC 2012 challenge was to detect the continuous dimensional emotions using audio-visual signals. In another notable work, Baltrusaitis et al. (2013) used continuous conditional random fields (CCRF) to jointly detect the emotional dimensions of AVEC 2012 continuous sub-challenge. This system achieved superior performance over SVR. For a comprehensive review of continuous emotion detection, we refer the reader to Gunes and Schuller (2013).

2.2 Approaches using physiological features

Various research works on emotion detection have already been explored using physiological features. Recognising human emotions induced by affective sounds through heart rate variability has been proposed in Nardelli et al. (2015). This article reported the method of recognition of emotional states revealed by affective sounds by means of estimates of autonomic nervous system (ANS) dynamics. The ANS dynamics was estimated through standard and nonlinear analysis of heart rate variability (HRV) exclusively, which was derived from the electrocardiogram (ECG). Inquisition was carried out on the synchronisation between breathing patterns and heart rate during emotional visual revealing in Valenza et al. (2012). Valenza et al. (2014) proposed human mood detection system using a wearable system. In this system, a comfortable t-shirt was used which was equipped with integrated fabric electrodes and sensors and was able to acquire ECG, respirogram and body posture information in order to detect a pattern of objective physiological parameters to support diagnosis. In another notable study, Partala and Surakka (2003) explored the variation of pupil size during and after emotional stimulation by external audio system. Aracena et al. (2015) depicted an emotion detection approach by creating signals of pupil size and gaze position observed during image viewing. Lanata et al. (2011) explored whether useful cues can be obtained from eye tracking and pupil size variation observed during image viewing at different arousal content obtaining from new wearable and wireless EGT. Frantzidis et al. (2008) designed an emotion detection system by fusing multi-modal physiological signals of the autonomic (skin conductance) and central nervous systems (EEG). Soleymani et al. (2016) developed a combined approach for emotion detection of video viewers' from electroencephalogram (EEG) signals and facial expressions. Happy and Routray (2015) presented a framework for emotion detection by applying appearance features of selected facial patches. In another study, Chakraborty et al. (2009) presented a fuzzy relational approach for human emotion recognition from facial expressions by applying external stimulus to excite specific emotions. In Zhang and Ji (2005), an active infra-red illumination along with Kalman filtering was used for accurate tracking of facial components. Martinez and Du (2012) proposed a facial expression-based emotion recognition model where the model consists of C distinct continuous spaces and multiple emotion categories can be recognised by linearly combining these C face spaces. According to this model, the major task for the classification of facial expressions of emotion is precise, detailed detection of facial landmarks rather than recognition. In another study (Smith and Windeatt, 2010), facial expression detection using filtered LBP features and by applying ECOC classifiers and Platt scaling was proposed. But, all the existing approaches of emotion detection applying physiological features require external equipments where these equipments are generally expensive and practically difficult to implement.

2.3 Text-based approaches

Apart from learning through psychological signals, some researchers proposed text-based model to detect human emotions. Truly speaking, the research on emotion recognition using text-based analysis is still in its initial stage. Generally, there are two common approaches to this task, namely a rule-based one and a machine-learning-based one. A rule-based system that tags emotions in news headlines was proposed and implemented by Chaumartin (2007). It computes word's sentiment polarity according to linguistic knowledge and predefined rules. Even though this system achieved a high accuracy, the recall was rather low. When it comes to the machine learning-based approach, Tan and Zhang (2008) explored four feature selection methods (MI, IG, CHI and DF) and five learning methods (centroid classifier, K-nearest neighbour, winnow classifier, naive Bayes and SVM) in an empirical study. The experiment results show that IG and SVM perform best. They also point out that classifiers severely depend on domains and topics. Tokuhisa et al. (2008) adopted the k -nearest-neighbour method and a two-step classification model. Based on a very big amount of data extracted from the web, this system significantly outperformed the baseline. Li et al. (2016) proposed hybrid neural networks based on biterm topic model (BTM), a variant of latent Dirichlet allocation, for social emotion detection. Li and Xu (2014) proposed a method for identifying emotions in microblog posts based on extracted cause events where the machine was trained using a single classifier. In another notable study, Ramakrishnan and Ibrahiem (2013) depicted an approach where a wide range of acoustic and linguistic features extracted for speech emotion recognition.

State of the art reveals that it is very difficult to predict the emotions of young generations, especially students from educational institutions. Not only that, nowadays many employees of industries also are suffering from depression due to work stress. They require proper psychiatric counselling to detect internal feelings. For smooth functioning of various organisations and institutions, mother-friendly atmosphere is highly desirable. The following studies depict the empirical study on work organisation and emotions:

Fregonese et al. (2018) proposed a conceptualisation and an operational measure for affective investment and symbolic motives at work. The work-symbolic motive scale (Work-SMS) score gave a general measure of affective investment. Another study (Jain et al., 2018) analysed the work/job related health disorders and difficulties which are faced by the faculty members of different educational institutions. This study considered emotional instability and spiritual health of employees. Springer and Bialowas (2018) proposed a method to detect the relationship among personality traits, core self-evaluation, emotional intelligence and positive employee outcomes (PO) in the context of specific demands of emotional labour. In this study research samples were consisted of 309 workers, out of which 170 workers performed emotional labour. The Polish adaptation of the NEO-FFI, the core self-evaluation scale and the INTE questionnaire were used to measure the level of emotional intelligence of employees. Blazovich et al. (2018) presented a study which found that mother-friendly firms experience better financial performance. This finding suggested the capital market value firms with 'mother friendly' attributes. It also found that enhancing work-life balance facilitates emotional well-being of employees, which is directly related with improved job performance and better financial performance of the company.

Most of the past researches using classification techniques employ empirical machine learning method. Contrary, it is almost impossible to solve recognition problem by only using empirical learning method without linguistic approach. Hence, the proposed approach solves the essential challenge of emotion recognition using a unique consolidated analysis of text and questionnaire-based data ensemble.

3 Theoretical background

Detection of human emotions has been carried out using two different classifiers – SVM and ANN, before combining the probability scores of these two classifiers using DST. The theoretical background of these two classifiers is discussed below.

3.1 SVM

In machine learning, SVMs are supervised learning models with associated learning algorithms that analyse data and recognise patterns, used for classification and regression analysis. Given a set of training examples, each marked for belonging to one of two categories, an SVM training algorithm builds a model that assigns new examples into one category or the other, making it a non-probabilistic binary linear classifier. An SVM model is a representation of the examples as points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible. New examples are then mapped into that same space and predicted to belong to a category based on which side of the gap they fall on. SVM has been used successfully for pattern recognition and regression tasks (Burges, 1998; Pal et al., 2010; Vapnik, 1995). SVM was originally defined for the problems of two-classes where it finds the optimal hyper-planes that maximise the margin between the positive and negative datasets of these classes. This hyper-plane is characterised by the normal vector, which is expressed as linear combination of the nearest examples of both classes, named support vectors. In order to extend SVM to solve multi-class pattern recognition problem, kernels technique is used.

More formally, a support vector machine constructs a hyper-plane or set of hyper-planes in a high- or infinite-dimensional space, which can be used for classification, regression, or other tasks. Intuitively, a good separation is achieved by the hyper-plane that has the largest distance to the nearest training-data point of any class (so-called functional *margin*), since in general the larger the *margin* the lower the generalisation error of the classifier.

Suppose, T_D is a training dataset consists of pairs (x_i, y_i) , $i = 1, 2, \dots, n$, $x_i \in R^n$ and $y_i \in (-1, 1)$, where x_i denotes input feature vector for i^{th} sample and y_i denotes the corresponding target value. For a given input pattern x , the decision function of an SVM binary classifier is

$$f(x) = \text{sign} \left(\sum_{i=1}^n y_i \alpha_i K(x, x_i) + b \right) \quad (1)$$

$$\text{sign}(u) = \begin{cases} 1, & \text{for } u > 0 \\ -1, & \text{for } u < 0 \end{cases} \quad (2)$$

b is the bias, α_i is the Lagrange multiplier and $K(x, x_i)$ is the kernel function.

The input feature vector x is mapped into higher dimensional feature space using the kernel function to make them linearly separable. Several kernel functions are used in SVM. Some of those kernel functions are Gaussian (radial basis function) kernel, polynomial kernel, linear kernel, etc. Studies (Scholkopf et al., 1997) have shown that RBF networks designed through support vector (SV) method can produce better recognition performances compared to those designed with traditional methodology for the same dataset.

3.2 ANN

Neural networks are a computational approach which is based on a large collection of neural units loosely modelling the way a biological brain solves problems with large clusters of biological neurons connected by axons. Each neural unit is connected with many others, and links can be enforcing or inhibitory in their effect on the activation state of connected neural units. Each individual neural unit may have a summation function which combines the values of all its inputs together. There may be a threshold function or limiting function on each connection and on the unit itself such that it must surpass it before it can propagate to other neurons. These systems are self-learning and trained rather than explicitly programmed and excel in areas where the solution or feature detection is difficult to express in a traditional computer program. Neural networks typically consist of multiple layers or a cube design, and the signal path traverses from front to back. Back propagation is where the forward stimulation is used to reset weights on the 'front' neural units and this is sometimes done in combination with training where the correct result is known. More modern networks are a free flowing in terms of stimulation and inhibition with connections interacting in a much more chaotic and complex fashion. Dynamic neural networks are the most advanced in that they dynamically can, based on rules, form new connections and even new neural units while disabling others. The goal of the neural network is to solve problems in the same way that the human brain would, although several neural networks are much more abstract. Modern neural network projects typically work with a few thousand to a few million neural units and millions of connections, which are still several orders of magnitude less complex than the human brain and closer to the computing power of a worm. New brain research often stimulates new patterns in neural networks. One new approach is using connections which span much further and link processing layers rather than always being localised to adjacent neurons. Other research being explored with the different types of signal over time that axons propagate which is more complex than simply on or off. Neural networks are based on real numbers, with the value of the core and of the axon typically being a representation between 0.0 and 1. An interesting facet of these systems is that they are unpredictable in their success with self learning. After training some become great problem solvers and others do not perform as well. In order to train them several thousand cycles of interaction typically occur. Like other machine learning methods – systems that learn from data – neural networks have been used to solve a wide variety of tasks, like computer vision and speech recognition, that are hard to solve using ordinary rule-based programming. Historically, the use of neural network models marked a directional shift in the late eighties from high-level (symbolic) artificial intelligence, characterised by expert systems with knowledge embodied in if-then rules,

to low-level (sub-symbolic) machine learning, characterised by knowledge embodied in the parameters of a dynamical system.

3.3 DST

DST or evidence theory is a general framework for dealing with uncertainty, with reference to other frameworks such as probability, possibility and imprecise probability theories. The theory involves combining evidence from different sources and arriving at a degree of belief that takes into account all the available evidences. The theory is specifically effective in combining multiple information sources involving incomplete, imprecise, biased and conflict knowledge. In Kessentini (2015) and Zadeh (1986), the authors have shown that DST can be employed to improve the accuracy rate and the reliability of an HMM-based handwriting recognition system. Similarly, the strategy can be further implemented on the combination of various classifiers. For this purpose, an evidential combination method is proposed to finely combine the probabilistic outputs of various classifiers.

A DST-based approach can be illustrated as follows – Let $\Omega = \{w_1, \dots, w_v\}$ be a finite set, also known as frame, formed by exclusive classes for each individual signature. A mass function μ is defined on the power set of Ω , represented as $P(\Omega)$, that maps onto $[0, 1]$ so that $\sum \mu(A) = 1$ where $A \subseteq \Omega$ and $\mu(\phi) = 0$. Then, a mass function is roughly a probability function defined on $P(\Omega)$ in lieu of Ω . It provides a broader description as the support of the function is enhanced: If $\|\Omega\|$ is the cardinality of Ω , then $P(\Omega)$ contains $2^{\|\Omega\|}$ elements (Kessentini, 2015).

The belief function *bel* is defined using (3).

$$bel(A) = \sum \mu(B); \forall A \subseteq \Omega, \text{ where } B \subseteq A, B \neq \phi \quad (3)$$

bel(A) refers to the probabilistic lower bound (i.e., all evidences that imply A). Similarly, the plausibility function *pl* is defined using (4).

$$pl(A) = \sum \mu(B); \forall A \subseteq \Omega, \text{ where } B \cap A \neq \phi \quad (4)$$

It refers to the probability of all the evidences that do not contradict A. Consequently, the difference between plausibility and belief, i.e., $pl(A) - bel(A)$ corresponds to the *imprecision* associated with subset A of Ω .

Two mass functions μ_1 and μ_2 based on the evidences of two independent sources can be combined into a consonant mass function using (5).

$$M(Z) = \frac{\sum_{A \cap B = Z} \mu_1(A) \times \mu_2(B)}{1 - \sum_{A \cap B = \phi} \mu_1(A) \times \mu_2(B)} \quad (5)$$

where $Z \neq \phi$, $Z \subseteq \Omega$, and A, B denotes two different sources. Evidential combination strategy (Kessentini, 2015) aims at combining the outputs of various classifiers, being utilised, in the best possible way. For this, the steps are:

- 1 building the frame
- 2 converting the probabilistic output of each of the Q classifiers into a mass function
- 3 computing the conjunctive combination of the Q mass functions
- 4 designing a decision function using pignistic transform.

4 Dataset development

4.1 Questionnaire approach

The dataset required for this approach has been created by us. Data have been collected via the Google cloud service, i.e., Google forms. The link was shared and a comprehensive set of people with a mix bag of males/females, of different age groups and of different professions were asked to fill the questionnaire. The responses of the emotional questionnaire have been collected from 400 different persons of varying age groups and occupations. The data collected are in the form of tabulated sheet which is further converted to tab separated sheet (.tsv) format so that data could be utilised for mapping into their corresponding mathematical value. Figure 1 shows a snapshot of data collected in questionnaire approach. In this figure, each row corresponds to the input from a person and each column defines a particular feature, e.g., sleep, heart rate, etc. It can also be noted from this figure that for a particular feature there can be more than one emotion related to it. In order to perform training these emotion strings were mapped to some numerical value. Mapping the string to the numerical value was required so that the SVM classifier could classify the values to a particular class level. So, the feedback of each person in response to the questionnaire contributes one feature vector for each emotion class. Thus, we have collected a total of 2,000 samples comprising all five emotion classes. These samples have been used only to train the system. For generating the testing dataset, 100 different persons from different age groups and occupations contributed their feedback against a similar form consisting of a set of questions and answers with options of 'yes', 'no' or 'cannot say'. Here, similarly like training dataset the feedback of each person in response to the questionnaire contributes one feature vector for each emotion class. Thus, we have collected a total of 500 samples to test our proposed system.

Figure 1 Snapshot of data collected in questionnaire approach (see online version for colours)

	A	B	C	D	E	F	G	H	I
1	When do you feel	When do you feel	When do you feel	When do you feel	When do you feel	When do you feel	When do you feel	When do you feel	When do you feel
2	Anger, Can't Say	Sad, Fear, Disgust	Sad, Fear, Anger, Disgust	Sad, Fear	Sad, Disgust	Can't Say	Can't Say	Joy, Anger	Sad, Fear, Disgust
3	Can't Say	Can't Say	Can't Say	Can't Say	Anger	Fear	Can't Say	Joy	Can't Say
4	Sad, Fear	Joy	Sad	Sad	Sad	Anger	Sad, Fear	Joy	Joy, Fear
5	Fear	Fear	Can't Say	Sad	Sad, Fear, Anger	Sad, Can't Say	Fear	Joy	Can't Say
6	Fear	Can't Say	Joy	Sad	Anger	Can't Say	Can't Say	Joy	Can't Say
7	Can't Say	Fear	Anger	Can't Say	Anger	Can't Say	Can't Say	Joy	Sad
8	Fear, Anger	Joy, Fear	Sad	Sad	Fear, Anger	Sad, Fear	Fear	Joy	Sad
9	Joy	Sad	Anger	Can't Say	Anger	Fear	Joy	Joy	Sad
10	Fear, Anger	Joy, Anger	Joy, Sad, Fear	Sad, Fear	Joy, Fear, Anger	Anger	Sad	Joy, Anger	Sad
11	Fear	Sad, Disgust	Sad, Disgust	Sad, Disgust	Can't Say	Can't Say	Can't Say	Sad, Disgust	Disgust
12	Fear, Anger	Fear	Sad	Fear	Sad, Anger	Fear, Anger, Disgust	Fear	Joy	Sad
13	Joy, Fear	Sad	Can't Say	Disgust	Can't Say	Can't Say	Can't Say	Joy, Anger	Can't Say
14	Joy, Sad	Can't Say	Sad	Sad	Joy, Anger	Fear	Fear	Joy	Sad, Anger
15	Joy	Joy, Sad	Sad	Sad	Joy, Sad, Fear	Can't Say	Can't Say	Joy, Anger	Joy
16	Fear	Sad, Disgust	Sad	Sad, Fear	Anger, Disgust	Joy	Fear	Joy, Anger, Disgust	Sad, Fear, Anger
17	Sad	Sad	Sad, Disgust	Joy	Fear	Fear, Anger	Can't Say	Joy	Sad, Anger
18	Fear	Sad	Sad	Sad, Fear	Sad, Fear	Fear	Sad, Fear	Joy	Fear
19	Fear	Fear	Can't Say	Fear	Joy, Fear	Fear	Joy	Joy, Anger	Sad
20	Joy, Sad	Sad, Anger	Can't Say	Sad, Anger	Sad, Anger	Anger	Can't Say	Joy, Anger	Joy
21	Fear	Sad	Joy	Joy	Anger	Fear	Can't Say	Can't Say	Sad
22	Fear, Anger	Sad, Fear	Joy	Sad	Sad, Disgust	Anger	Can't Say	Joy	Sad
23	Can't Say	Anger	Anger	Fear	Fear	Fear	Fear	Joy	Fear
24	Can't Say	Joy	Joy	Can't Say	Fear	Anger	Sad	Joy	Sad
25	Fear	Can't Say	Sad	Can't Say	Joy	Can't Say	Can't Say	Sad	Anger
26	Fear	Joy, Sad, Fear	Sad, Fear, Anger	Fear	Sad	Fear	Can't Say	Joy	Can't Say

4.2 Text analysis-based approach

In text analysis-based approach, training data have been collected from one publicly available dataset – ISEAR. Each item in ISEAR dataset is a blog written by somebody when he/she was feeling a particular emotion. Each blog is labeled with a particular emotion class. The snapshot of few blog items from ISEAR dataset is shown in Figure 2. This dataset contains a total of 7,666 items. Out of these, 2,000 items have been considered as training samples to train the present system. These 2,000 samples have been collected keeping in mind the uniform representation of samples from each class. For generating the testing dataset, 100 different persons from different age groups and educational backgrounds contributed their feedback by writing one blog which is mapped to all five emotional class labels.

Figure 2 Snapshot of data collected in text analysis-based approach

[anger]	When I was talking to HIM at a party for the first time in a long while and a friend came and interrupted us and HE left.
[anger]	When a car is overtaking another and I am forced to drive off the road.
[sad]	When I lost the person who meant the most to me.
[sad]	When my friends did not ask me to go to a New Year's party with them.
[joy]	When I got a letter offering me the Summer job that I had applied for.
[joy]	After my girlfriend had taken her exam we went to her parent's place.
[disgust]	When I saw all the very drunk kids (13-14 years old) in town on Walpurgis night.
[disgust]	When I found a bristle in the liver paste tube.
[fear]	When I was involved in a traffic accident.
[fear]	When one has arranged to meet someone and that person arrives late.

4.3 Combination of questionnaire and text analysis-based approaches

For combining the feature vectors of questionnaire and text analysis-based approaches, a total of 2,000 samples have been considered for training from each of the two datasets – our own dataset on questionnaire approach and ISEAR dataset for text analysis-based approach. These many samples from questionnaire approach have been collected from 400 different persons of varying age groups and occupations, whereas out of 7,666 samples available in the ISEAR dataset, a total of 2,000 samples have been considered keeping in mind the uniform representation of samples from each class. To carry out the testing, a total of 500 aforesaid samples of questionnaire and text analysis-based approaches have been combined.

The detailed statistics of datasets used in this work are shown in Table 1.

Table 1 Details of the datasets used in our experiment

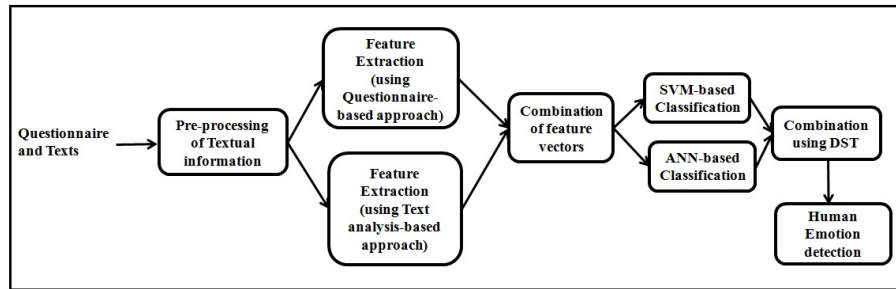
<i>Approach</i>	<i>Training + testing samples</i>	<i>Train – test dataset</i>
Questionnaire-based	2000 + 500	Own – own
Text analysis-based	2000 + 500	ISEAR (Public) – own
Combined approach	2000 + 500	Own + ISEAR – own

5 Proposed approach

In this work, we have analysed a culmination of questionnaire and text analysis-based approaches to determine the emotional state of a person. Features have been extracted

separately from both questionnaire and text analysis-based approaches. Next, features obtained from these two approaches are combined to generate the final feature vector. These feature vectors are studied in both SVM and ANN-based platforms to determine the emotional state of a person. Finally, to enhance the performance of the system, the probability scores of SVM and ANN have been combined using DST. The detailed block diagram of the proposed framework is shown in Figure 3.

Figure 3 Detailed block diagram of the proposed framework



5.1 Questionnaire-based approach

5.1.1 Feature extraction

There are several features/characteristics that correspond to a particular emotion. Upon research and proper consultation with a practicing psychiatrist a set of most suitable features has been created corresponding to each emotion. Thus, 30 most suitable features came upon covering aforesaid five emotions. Those 30 features are shown in Table 2.

Table 2 Features used in questionnaire-based approach

Feature number	Feature name	Feature number	Feature name
1	Increase in heart rate	16	Pen down your thoughts
2	Lack of sleep	17	Feel like harming yourself
3	Lack of appetite	18	Talk less
4	Speak in low voice (amplitude)	19	Listen to soft music
5	Trouble in concentrating	20	Unconscious of surroundings
6	Voice shiver	21	Perspire (sweat) more
7	Biting nails	22	Feel optimistic
8	Feel more energetic	23	Use slang language while speaking
9	Talk less	24	Breath rapidly/unevenly
10	Cry	25	Grind your teeth
11	Need to share your feeling with other	26	Feeling of revenge (destructive)
12	Avoid eye contact with others	27	Want to hug someone
13	Cannot act rationally	28	Rub your palms
14	Wish to be alone	29	Blame yourself for the situation
15	Feel discomfort	30	Feel confident

Mapping the emotion string to numerical values is done on the basis of data collection format. Each of the features starting from heart rate to the last feature in the row is assigned feature value from 1 to 30 based on their order of occurrence in the data collection format. For example, 'increase in heart rate' is assigned the value of 1. Each emotion string is assigned any class label between 1 and 5 depending on the emotion class in which it belongs. The class label is assigned as follows: anger – 1, sad – 2, joy – 3, disgust – 4 and fear – 5. The following equation has been used to deduce the numerical value for each particular entry in the cell of tabulated sheet that corresponds to that particular feature. The equation is:

$$C(i) = (V_i)/(F_j * H) \quad (6)$$

where $V_i = i^{\text{th}}$ class value, $F_j = j^{\text{th}}$ feature value, $i = 1, \dots, 5$, $j = 1, \dots, 30$ and H is any prime number which is used to get a distinct numeric value for each feature. Equation (6) has been used as the mapping function where the value of H has been chosen as 19, because it provides the best result. This function has been used to generate the training and testing datasets. The feature values have been generated based on the following criteria: if for a particular cell, the value entered by the subject is 'cannot say', i.e., the person is not sure about under which emotion that particular feature should lie, then it is assigned the value -0.10 . For a particular cell that has at most five emotions, the numerical value can be calculated for each emotion using the mapping equation (6). If any emotion does not belong to a particular cell then the value 0.00 is assigned as the mapping value for that particular feature of that emotion. Thus, in this approach, all the collected samples are broken down in such a way that the feedback of each person creates five separate feature vectors corresponding to five emotion classes. Each feature vector will be a result of presence/non-presence of a particular emotion of each sample for each question. Thus, a 30-dimensional feature vector is generated for each class from each sample in this approach.

5.2 Text analysis-based approach

5.2.1 Pre-processing

For both training and testing datasets of text, first of all, we need to carry out some pre-processing tasks on the collected data. Initially, each sample is converted to lowercase and the characters other than $a-z$ are removed from the sentences as they do not influence the emotion or state of a person ('well it does'). Then, we have eliminated the 'stop-word' (e.g., 'the', 'that' and lexical words) from the sample and at last checked the negation in the sentence. If the word 'not' comes with a verb, adjective or adverb then it has been combined with the word for further consideration, otherwise the negation is removed as again it is not going to influence the sentence for the emotion.

5.2.2 Feature extraction

The basic emotions are the only key features as the text consists of the basic emotions whose values will be the probabilities of the emotion state in the sentence. These basic emotions are *anger*, *sad*, *joy*, *disgust* and *fear*.

In this approach, the text is converted into tokens applying Natural Language Toolkit (NLTK) word tokeniser. Then, each token is added with the POS tag using the NLTK POS tagger. Among the tags, only adjective, verb or adverb tokens are selected as these are directly related to the emotion of human being. Now, using the National Research Council (NRC), Canada word-emotion lexicon, we find the token(s) corresponding to each emotion. Finally, we calculate the probability of each emotion in a particular blog using the following equation:

$$C(i) = (N_i) * \delta / (N) \quad (7)$$

where N_i = number of tokens related to i^{th} emotion in the blog, N = total number of tokens in the entire dataset, δ is a constant used to prevent generating very small fractional number of $C(i)$, $i = 1, \dots, 5$.

The feature values have been generated based on the following criteria: if any blog contains two tokens related to a particular emotion class, say, 'fear', 1 token related to another token class, say, 'sad' and no token related to other classes then probability of 'fear' class is $(2 * \delta) / N$, probability of 'sad' class is $(1 * \delta) / N$ and probabilities of other emotion classes are 0. Thus, each sample contributes a 1-dimensional feature vector for each emotion class in this approach.

5.3 Generation of feature vector by combining both approaches

We have tested our system only by combining the feature vectors obtained from each of the two approaches – questionnaire and text analysis-based to evaluate the performance of the system. After combining, a 31-dimensionality feature vector has been generated for each of the training and testing samples – 30 features from questionnaire-based approach and 1 feature from text analysis-based approach.

5.4 Combining the probability scores of two classifiers

After generating the final feature vector by combining questionnaire and text analysis-based approaches, each feature vector is studied in both SVM and ANN-based platforms to detect the emotional state of a person. Both SVM and ANN classifiers return one probability score corresponding to each emotion class and as a result a total of ten probability scores are obtained for five different emotion classes from these two classifiers. Thus, the following two sets of probability scores are generated from these two classifiers:

$$\alpha = \{SAnger, SSad, SJoy, SDisgust, SFear\} \quad (8)$$

$$\beta = \{NAnger, NSad, NJoy, NDisgust, NFear\} \quad (9)$$

Here, α denotes the set of probability scores obtained from SVM classifier, whereas β denotes the set of probabilities gained from ANN classifier.

Algorithm 1 Combining the probability scores of SVM and ANN classifiers using DST**Input:** Probability scores of emotional classes obtained from sSVM and ANN classifiers

Step 1:

Preparation of training dataset

- I. Train_Mat1 size $600 \times 37 = 600$ training samples * 37 features *and*
where 37 features = 30 features from questionnaire + 7 features from text approach
- II. Train_Mat2 size $600 \times 1 = 600$ class labels

Preparation of testing dataset

- I. Test_Mat1 size $46 \times 37 = 46$ testing samples * 37 features *and*
where 37 features = 30 features from questionnaire + 7 features from text approach
- II. Test_Mat2 size $46 \times 1 = 46$ class labels

Step 2: Model Train_M1 = SVM(Train_Mat1, Train_Mat2)

Model Train_M2 = NN(Train_Mat1, Train_Mat2)

Step 3: Test_M1 = Train_M1(Test_Mat1)

Test_M2(Test_Mat1)

Step 4: SVN_Accuracy = Accuracy(Test_M1, Test_Mat2)

NN_Accuracy = Accuracy(Test_M2, Test_Mat2)

Step 5: Final_Accuracy = DST(Test_M1, Test_M2, Test_Mat2)

DST has been used to combine the probability scores of these two classifiers. Algorithm 1 shows the major steps of combining the probability scores of SVM and ANN classifiers using DST. The aim is to combine the outputs of SVM and ANN classifiers in the best way. To accomplish this, it is required to convert the probabilistic output of each of these two classifiers into a mass function, compute the conjunctive combination of the mass functions, and design a decision function. Initially, the probabilistic output of each of the two classifiers is converted into a mass function. The inverse pignistic transform converts an initial probability distribution p into a consonant mass function. The resulting consonant mass function is computed in the following way: let p_i be the probability value of a particular emotional approach set Ω corresponding to a particular emotion. First, the elements of Ω are ranked in decreasing order of probabilities as defined in (10),

$$p(e_1) > \dots > p(e_{|\Omega|}). \quad (10)$$

Next, mass function μ is described using (11) and (12),

$$\mu(\{e_1, e_2, \dots, e_{|\Omega|}\}) = \mu(\Omega) = |\Omega| \times p(e_{|\Omega|}) \quad (11)$$

$$\forall i < |\Omega|, \mu(\{e_1, e_2, \dots, e_i\}) = i \times [p(e_i) - p(e_{i+1})] \quad (12)$$

In our framework, six emotional classes have been considered, hence, $|\Omega| = 6$. $\mu_1(e_i)$, $\mu_1(e_i, e_{i+1})$, $\mu_1(e_i, e_{i+1}, e_{i+2})$, $\mu_1(e_i, e_{i+1}, e_{i+2}, e_{i+3})$, $\mu_1(e_i, e_{i+1}, e_{i+2}, e_{i+3}, e_{i+4})$ and $\mu_1(e_i, \dots, e_{i+5})$ have been obtained from the resultant probability set of SVM classifier. Here, each subset of μ_1 is represented by X . Similarly, $\mu_2(e_i)$, $\mu_2(e_i, e_{i+1})$, $\mu_2(e_i, e_{i+1}, e_{i+2})$, $\mu_2(e_i, e_{i+1}, e_{i+2}, e_{i+3})$, $\mu_2(e_i, e_{i+1}, e_{i+2}, e_{i+3}, e_{i+4})$

and $\mu_2(e_i, e_{i+1}, e_{i+2}, e_{i+3}, e_{i+4}, e_{i+5})$ have been obtained from the resultant probability set of ANN classifier. Here, each subset of μ_2 is represented by Y . The above mass functions are combined using (13),

$$M(A) = \frac{\sum_{X \cap Y = A} \mu_1(X) \times \mu_2(Y)}{1 - \sum_{X \cap Y = \phi} \mu_1(X) \times \mu_2(Y)} \quad (13)$$

where, $A \neq \phi$ and $A \subseteq \Omega$.

For decision making, *belief*, *plausibility* and *conflict* are computed for each emotion class E using (14)–(16),

$$\text{belief}(E) = \sum_{A \subseteq E} M(A) \quad (14)$$

$$\text{plausibility}(E) = \sum_{A \cap E \neq \phi} M(A) \quad (15)$$

$$\text{conflict}(E) = \text{plausibility}(E) - \text{belief}(E) \quad (16)$$

The calculated *conflict* for each of the emotions is compared with the threshold value T , whose value is in $[0, 1]$. This threshold value is decided based on the experiment and compared with the conflict values for the most suitable emotion. The emotions E , satisfying the relation (17),

$$\text{conflict}(E) \leq T \quad (17)$$

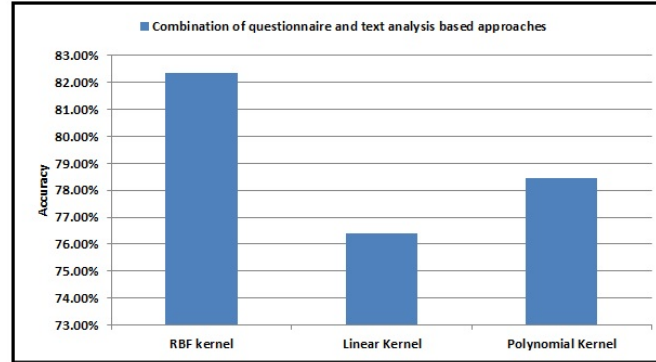
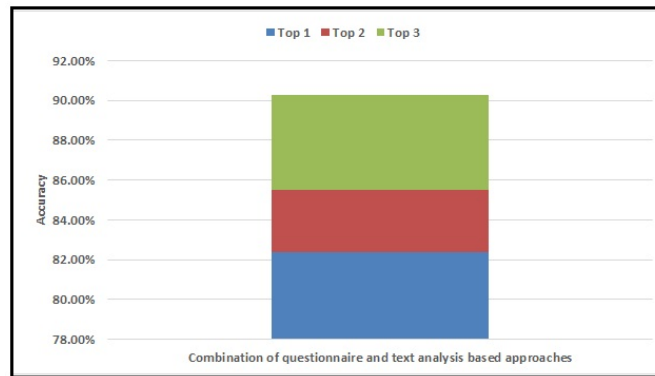
are the *candidate emotion(s)* to be accepted. Among these candidate emotions, the emotion with the lowest conflict value is accepted as final emotional state of the person and others are rejected. If it happens that none of the *conflict*(E) values is satisfying the relation (17), then we will not accept any of the emotions as relevant and thus, reject all the emotions E , for the result.

6 Performance analysis

The performance of the proposed system has been evaluated only after combining the feature vectors of questionnaire and text analysis-based approaches using both SVM and ANN. Next, to enhance the performance of the system, the probability scores of SVM and ANN classifiers have been combined using DST.

6.1 Results using SVM

The performance of the proposed system has been tested using various SVM kernels. Detailed performance analysis of the proposed system, using SVM is shown in Figure 4. It can be seen from Figure 4 that radial basis function (RBF) kernel has provided best emotion detection performance among others. Figure 5 shows the performance of the proposed system using SVM when top three choices are considered.

Figure 4 Performances of the proposed system using SVM (see online version for colours)**Figure 5** Performance of the proposed system using SVM considering top three choices (see online version for colours)

6.2 Results using ANN

To test the performance of the proposed system using ANN, experiments have been carried out by varying the number of hidden layers from 1 to 3 and varying the number of neurons per hidden layer. Detailed performance analysis of the proposed system using ANN is shown in Table 3. It can be noted that, the proposed system has shown better performance using ANN compared to SVM.

Table 3 Performances of the proposed system using ANN

<i>Hidden layer</i>	<i>Number of neurons</i>	<i>Accuracy</i>
2	32-32	77.21%
2	80-80	80.43%
3	32-50-70	83.38%
3	50-70-70	84.92%

6.3 Results using DST

After evaluating the performances of the proposed system separately using SVM and ANN, the probability scores of SVM and ANN have been combined using DST for enhancing the performance of the system. Table 4 shows the performance of the proposed system after combining the probability scores of SVM and ANN using DST. These probability scores are obtained from the combined feature vector of both approaches.

Table 4 Performance of the proposed system after combining the probability scores of SVM and ANN using DST

<i>Approach</i>	<i>Accuracy</i>
Combination of questionnaire and text analysis-based approaches	93.28%

To measure the performance of the proposed system when DST is applied, we use *precision*, *recall* and *F1-score* as performance measurement parameters using (18)–(20).

$$Precision = \frac{TP}{TP+FP} \quad (18)$$

$$Recall = \frac{TP}{TP+FN} \quad (19)$$

$$F1-Score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (20)$$

The parameters used in the above equations [equations (18) and (19)] are described in Table 5.

Table 5 Parameters used for computation of precision and recall

<i>Parameters</i>	<i>Full form</i>	<i>Description</i>
TP	True positive	Correctly predicted positive values (values of both actual and predicted classes are ‘yes’)
FP	False positive	The value of actual class is ‘no’ and predicted class is ‘yes’
TN	True negative	Correctly predicted negative values (values of both actual and predicted classes are ‘no’)
FN	False negative	Here actual class is ‘yes’ and predicted class is ‘no’

Figures 6, 7 and 8 show the *precision*, *recall* and *F1-score*, respectively of the proposed approach using SVM, ANN and DST.

Figure 6 *Precision* of the combined approach using SVM, ANN and DST (see online version for colours)

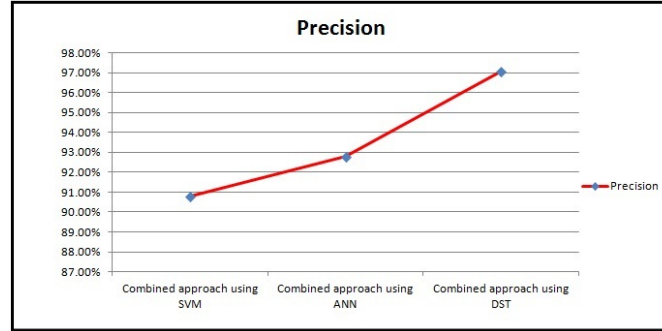


Figure 7 *Recall* of the combined approach using SVM, ANN and DST (see online version for colours)

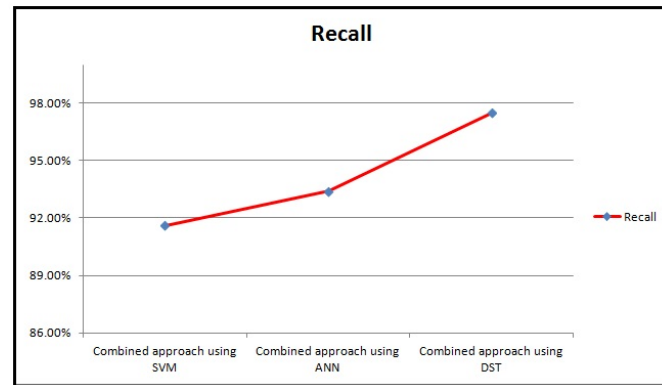
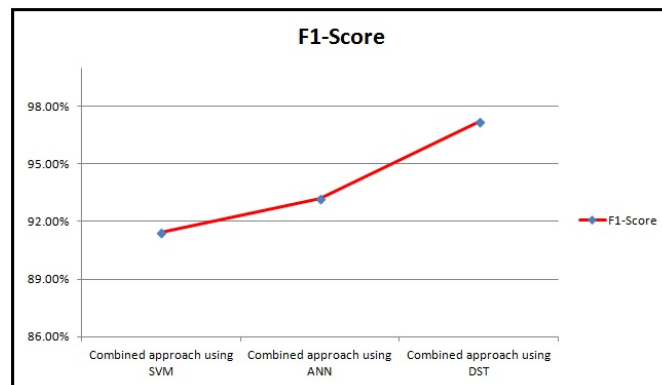
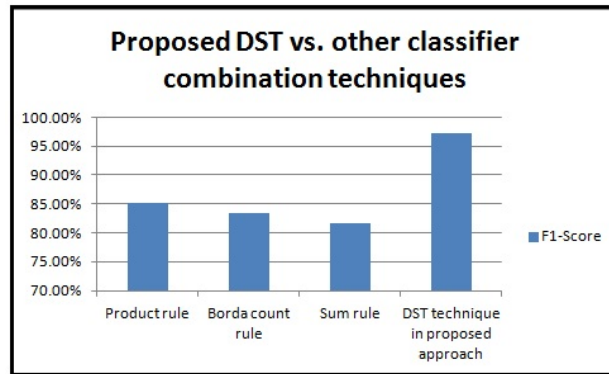


Figure 8 *F1-Score* of the combined approach using SVM, ANN and DST (see online version for colours)



The proposed DST technique-based result has also been compared with other existing classifier combination techniques such as sum, product and Borda count rules. The comparisons are shown in Figure 9, where the proposed technique-based result outperforms the other techniques.

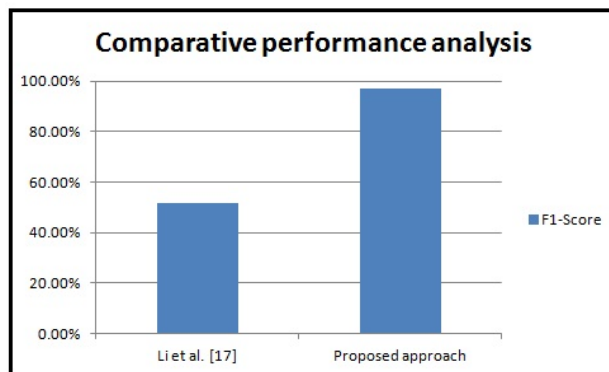
Figure 9 Comparative performance analysis of various classifier combination techniques in the proposed approach (see online version for colours)



6.4 Comparative performance analysis

In order to evaluate the effectiveness of the proposed approach towards human emotion detection, it is vital to carry out a comparative analysis with already existing benchmark approaches based on text-analysis. We have used ISEAR, a public dataset for analysing text-based approach. Some of the existing text-based approaches have been tested on different datasets and hence cannot be compared directly with the proposed approach. So, we have analysed the comparative performance of the proposed approach only with the study mentioned in Li et al. (2016). Figure 10 shows the desired comparative analysis.

Figure 10 Comparative performance analysis with the approach of Li et al. (2016) (see online version for colours)



7 Conclusions

Human emotion detection is a field that has tremendous potentials and is being explored. Emotion plays an important role in human communication such as different tweets and blogs written on social websites. Most of the approaches for emotion detection such as facial recognition, brain signal analysis, etc. are expensive and complex. A sound, reliable and simple emotion detection approach would open the doors to solve the problems of multitude in this society. This article has explored a novel emotion detection model by applying the combination of questionnaire and text analysis-based approaches and then combining probability scores of different classifiers (SVM and ANN) using DST to determine the emotional state of the subject with a certain accuracy. This type of emotion detection system can assist the society to detect human emotions, especially of younger generations in order to prevent them to commit suicide, apart from other societal benefits like human-computer interaction. The novelty of this technique lies in its simplicity and feasibility compared to other existing approaches.

Acknowledgements

We would like to acknowledge support for this research exploration from Mrs. Saroj Verma, Psychological Counsellor, National Institute of Technology Patna, India.

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